

Pankaj Kumar

AI Backend Engineer | System Architect

Punjab, India | +91-7717565570 | peter97986266@gmail.com | [LinkedIn](#) | [GitHub](#) | [X.com](#)

Summary

AI Backend Engineer experienced in building AI systems using RAG and multi-agent architectures. Strong expertise in Claude Code CLI for agent-driven coding and developer automation. agent workflows, and retrieval pipelines for document analysis

Technical Skills

- **Languages:** Python 3.11, SQL, [gRPC \(Protocol Buffers\)](#).
- **Core ML & DL:** PyTorch, Scikit-learn, Neural Network Architectures (Transformers, CNNs, RNNs), Model Fine-Tuning (LoRA, QLoRA, PEFT, RLHF)
- **Agentic AI:** LangGraph, PydanticAI, CrewAI, AI Evaluation (LLM-as-a-Judge, RAGAS, BLEU/ROUGE)
- **RAG:** GraphRAG, Hybrid Retrieval Pipelines, [DSPy \(Prompting\)](#)
- **LLM Serving:** vLLM, OpenAI, Google Gemini, VertexAI, Moonshot, Groq, LiteLLM
- **Databases:** Qdrant (Vector), Neo4j (Graph), Elasticsearch, Redis, Data Ingestion & Processing Pipelines
- **Infrastructure:** Docker, GitLab CI/CD, MinIO, Celery, nginx
- **Observability:** Structlog, Opik Observability

Work Experience

AI Backend Engineering Intern | Confidential (Legal Tech) / [luminova.ai](#) - Remote

Worked as a backend engineer on a AI platform for legal document analysis.

- Built and owned a FastAPI backend supporting multiple AI pipelines (Q&A, comparison, reports).
- Built hybrid RAG pipelines using Qdrant (vector) and Neo4j (graph).
- Improved performance with batching, tuning, and vector quantisation (up to 4× memory reduction).

Key Projects

1. [vibeVoice / Parakeet Flow v2](#) — Local Push-to-Talk Voice-to-Text Tool

- Built a local desktop voice-to-text app for macOS that transcribes speech and auto-pastes text into the active application.
- Implemented a streaming architecture with ear.py capturing mic audio via a PyAudio callback and sending PCM chunks to brain.py over a Unix socket
- Added a Parakeet-TDT backend through sherpa-onnx for optional high-accuracy transcription.
- Added runtime model switching through terminal hotkeys, supporting 10 selectable models from tiny.en through Parakeet-TDT variants.

2. [vLLM Model Hosting & Inference Platform](#) Self-Hosted | Built a complete LLM orchestration

platform featuring a Next.js dashboard and automated deployment scripts for CPU (Ollama) and GPU (vLLM) inference.

- Developed a management dashboard using Next.js 14 and shadcn/ui for model control, system health monitoring, and interactive chat.
- Engineered deployment scripts with auto-detection for hardware (CPU/GPU) and model size, handling automatic quantization and systemd service creation.

Education

Bachelor of Science (B.Sc.) - Mathematics

DAV College, Abohar (Panjab University) | Punjab, India -INROLE

- **Relevant Coursework:** Calculus, Linear Algebra, Probability & Statistics.

Achievements & Highlights

- Recognised for Agentic AI and RAG systems, including multi-agent orchestration and hybrid retrieval design.
- Designed and shipped a AI backend handling complex legal workflows.
- Demonstrated strong systems thinking across API design, async processing, and LLM orchestration.
- Active GitHub contributor integrating AI research concepts into real backend systems.